

PAPER_227: Tapestry of Blazing Starbirth (NGC 2014/2020, LMC) — MUGE with Gas-Accreting Mass $M(t)$ and Stellar Wind Ram Pressure

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Session: 58 (grokshare8d951e12.txt extraction — Doc 4)

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Author: Daniel T. Murphy **Framework:** UQFF v4.8 (Star-Magic) **Session:** 58 (grokshare8d951e12.txt extraction — Doc 4) **Date:** March 2026 **Classification:** Novel MUGE — Gas-Ratio Amplitude $M(t)$ + Stellar Wind Ram Pressure **Status:** Proof-Quality Whitepaper

Abstract

The Tapestry of Blazing Starbirth (NGC 2014 and NGC 2020) in the Large Magellanic Cloud is modelled using a 9-term MUGE incorporating two novel mathematical methods: (1) time-varying stellar mass $M(t) = M_{\text{init}}(1 + M_{\text{dot_factor}} \cdot e^{-t/\tau_{\text{SF}}})$ where the amplitude $M_{\text{dot_factor}} = M_{\text{gas}}/M_{\text{init}} \approx 41.7$ encodes the gas-to-stellar mass ratio, and (2) stellar wind ram-pressure acceleration $a_{\text{wind}} = \rho_{\text{wind}} v_{\text{wind}}^2 / \rho_{\text{fluid}}$. At canonical parameters $a_{\text{wind}} \approx 4 \times 10^3 \text{ m/s}^2$ dominates most other MUGE terms during active O/B-star formation.

1. Physical System

NGC 2014 and NGC 2020 are two giant H II regions in the Large Magellanic Cloud, separated by ~300 ly, forming the "Tapestry of Blazing Starbirth" (HST WFC3 image, 2020). Key parameters:

Parameter	Value
Distance	~160,000 ly (LMC)
M_{init}	240 $M_{\odot}$ (stellar; young massive stars)
M_{gas}	10{,}000 $M_{\odot}$ (surrounding gas)
$M_{\text{dot_factor}}$	$M_{\text{gas}}/M_{\text{init}} = 41.7$
r	10 ly
B	1 μT
z	LMC ≈ -0.0005 (Milky Way satellite)
τ_{SF}	5 Myr

2. Novel Equations

2.1 Gas-Ratio-Amplitude Mass Growth

$$M(t) = M_{\text{init}} \left(1 + \frac{M_{\text{gas}}}{M_{\text{init}}} e^{-t/\tau_{\text{SF}}} \right)$$

The amplitude $M_{\text{gas}}/M_{\text{init}}$ is the gas-to-stellar mass ratio, here ≈ 41.7 . This captures the rapid initial mass increase as gas accretes onto young stellar objects. By $t = 5\tau_{\text{SF}}$, M returns to M_{init} .

2.2 Stellar Wind Ram Pressure Acceleration

$$a_{\text{wind}} = \frac{\rho_{\text{wind}} \cdot v_{\text{wind}}^2}{\rho_{\text{fluid}}}$$

Parameter	Value
ρ_{wind}	10^{-21} kg/m^3 (O/B-star wind)
v_{wind}	$2000 \text{ km/s} = 2 \times 10^6 \text{ m/s}$
ρ_{fluid}	10^{-21} kg/m^3 (ambient medium)

At these parameters: $a_{\text{wind}} = (10^{-21} \times 4 \times 10^{12}) / 10^{-21} = 4 \times 10^{12} \text{ m/s}^2$.

3. Context Within MUGE Stellar Wind Family

System	ρ_{wind} (kg/m ³)	v_{wind} (km/s)	ρ_{fluid} (kg/m ³)
Tapestry LMC	10^{-21}	2000	10^{-21}
Westerlund 2	10^{-20}	2000	10^{-21}
NGC 1792 (SN)	10^{-21}	2000	10^{-21}

Westerlund 2 has ρ_{wind} $10\times$ higher, distinguishing it as the denser OB-supergiant environment.

4. Simulation Parameters

Parameter	Value
$t_{\text{canonical}}$	1 Myr
dt	0.01 Myr
τ_{SF}	5 Myr
E_0 erosion	No erosion term (use PAPER_229 for Pillars)

5. Calculator Class

```
class StarbirthTapestryLMCUQFFCalculator(_CP3Calculator):
    """PAPER_227: NGC 2014/2020 LMC Tapestry – 9-term MUGE, gas-ratio M(t), stellar wind"""
    # Session 58 – grok_share_8d951e12.txt Doc 4
```

Located in CondensedPhysics3.py (Session 58).

6. Conclusion

The Tapestry of Blazing Starbirth introduces two novel MUGE methods: gas-ratio-amplitude mass evolution and stellar wind ram-pressure acceleration. The a_{wind} term, at $\rho_{\text{wind}} = \rho_{\text{fluid}}$,

evaluates to v^2_{wind} — numerically dominant during the O/B-star-active phase. This establishes a parametric wind family within the LMC/Milky Way comparison (see PAPER_228 for the 10× denser Westerlund 2 case).

Source: grokshare8d951e12.txt — Doc 4 (Tapestry Starbirth LMC MUGE)

UQFF computed: UQFF magnetic Jeans correction factor $[SSq]B/(8p\mu cs) = 5.7e-1 \text{ } 1.3e-9 = 7.4e-10$; *Jeans mass deviation from standard* = $7.4e-10 \text{ } MJ$.